

THE CORE

C++/CUDA kernel fundamentals - gates, ladder, and session charter

v2 - 25 September 2026, San Diego.

“So many CPU threads and GPU cores under my control: silicon beasts in pools of large sizes - I get to task them, see them swim, and instrument how well they do. And CUDA/C++ teaches me to control these powerful processors.”

- owner's ignition packet, 25 September 2026

“Only two threads: blog-post research writing as play, C++/CUDA kernel as work.”

- owner's ruling, 22 September 2026

1 Charter and scope

The core is one capability. Take a single-GPU operation from a PyTorch reference to a correct CUDA C++ kernel; profile it; drive it toward a known fraction of the hardware limit; explain every remaining percent. C++ is learned *inside* this loop, never before it - kernel C++ is a dialect, repaired by typing daily, not by reading first.

Two threads only, until the gates fall. WORK: C++/CUDA kernel reps, in the morning slots, day opens here. PLAY: blog-post research writing, recreational and time-upper-bounded; architecture reading lives only inside this lane. Everything else on the board is out of scope for this document and this phase.

Design principles of this document. Gates, not dates - nothing here carries a deadline, a streak, a fraction, or one syllable of reproach about any gap. Completion is defined by receipts only. Gaps end by returning; the ledger records resumptions, never lapses. A thing conceived is not a thing constructed until its verification artifact exists.

The keeper's law. Every instrument in this lane faces the silicon. The beasts are the Oryx Pro (oryp11) now and the Thelio Mega when it lands; the pools are their memory; kernels are how the keeper tasks them. Nsight is a beast's telemetry; the compiler is the gate to the pool; the harness is the feeding record. The keeper reads every gauge; no gauge ever reads the keeper. Receipts attest artifacts and their build conditions - never verdicts on the author - and the measurement's object never rotates inward. This lane is escort terrain, lawfully measured; the sanctum stays elsewhere, unmeasured, by standing law.

The firewall. The exam register has no jurisdiction here: comparison to other people has no representation anywhere in the lane; the market's calendar lives in the career documents and ends at this field's door; the workplace remains untouchable entire.

2 The four gates

The core closes when all four artifact receipts exist. Nothing else counts toward it - and nothing about it counts toward the owner.

G1 - Extempore correctness. A correct reduction or row-softmax kernel exists, born in one sitting from a blank editor: no references open, its own test harness. The references-closed condition is a fact of the artifact's build.

Receipt: a dated repo commit containing kernel + harness, with a journal line attesting the references-closed condition.

G2 - Tiled SGEMM, gap accounted. A tiled SGEMM exists at a self-declared fraction of cuBLAS on the machine at hand, with every remaining percent of the gap named from Nsight Compute evidence (limiter, occupancy, memory behavior). The fraction is declared in the writeup, not negotiated afterward; the gap belongs to the kernel.

Receipt: ncu report + writeup naming the declared fraction and each residual percent's cause.

G3 - A fused ML op, both directions. An RMSNorm-class fused kernel exists, forward and backward, passing PyTorch gradcheck, benchmarked against the framework baseline.

Receipt: gradcheck pass log + benchmark table inside the writeup.

G4 - Read a stranger's kernel. Preregistered-oracle protocol: a public kernel whose author published their own optimization analysis, profiled independently; a dated limiter verdict committed before the author's analysis is read; then the comparison. The verdict document is the artifact; the match/mismatch note is data about the verdict.

Receipt: the committed verdict, the author's analysis, and the match/mismatch note.

3 Bring-up: two evenings, capped

Toolchain on the Oryx Pro (Pop!_OS). If this exceeds two evenings, the setup is the bug - debug it as one; swim time is not spent here.

- Driver check: `nvidia-smi` reports the GPU.
- CUDA toolkit installed; `nvcc --version` answers; one hello-kernel compiles and runs.
- Nsight Compute (ncu) and Nsight Systems (nsys) installed and launch.
- Python venv with CUDA PyTorch, for reference implementations and correctness harnesses (harness ports to C++ after G1).
- One git repo initialized: kernels, harnesses, journal, writeup drafts. The MacBook is the reading-and-notes machine only.

Measurement discipline (mobile silicon). AC power, performance profile, warmup runs, median-of-N - laptop GPUs thermally throttle. Until the Thelio lands, all numbers are relative: fraction of this card's own peak bandwidth, fraction of cuBLAS on this card - never absolute. The Ada mobile part lacks the datacenter features; nothing in the core depends on them. Absolute speed-of-light work and the tensor-core (WMMA) rung move to the Thelio when it arrives.

4 The shelf - one lawful door per book

Spine (in-session): PMPP 5th ed. - Hwu, Kirk, El Hajj, *Programming Massively Parallel Processors* - read just-in-time inside the session: the chapter that serves the current rung, forty minutes at most; the rest is typing. The ladder's pairings follow the 5th edition's numbering.

In-loop references (open on any question, any time): Josuttis, *C++20 - The Complete Guide*, for language questions arising from the kernel or harness at hand; the CUDA C++ Programming Guide (docs.nvidia.com); cppreference.com.

Behind-the-rep tier, under one law - a book's chapter on a kernel opens only after that kernel exists in the repo: Wilt, *The CUDA Handbook* - deep on the memory system, streams, and events; its hardware specifics are Kepler-era, so it is read for mechanisms, never for numbers. Ansorge, *Programming in Parallel with CUDA* - a second worked treatment of the same rungs; calibration after the kernel runs, temptation before.

Event-gated: Swidzinski, *Modern CMake for C++* - enters at the harness port after G1, nowhere earlier. Kerrisk, *The Linux Programming Interface* - shelved except targeted lookups (pinned memory, mmap) when ncu raises the question.

Other jurisdictions, so the shelf stays quiet: Scott, *Programming Language Pragmatics*, and Nystrom, *Crafting Interpreters*, belong to the play thread. Hellmann and Phillips (Python) are harness-side reference only. Stroustrup's *TC++PL* (C++11) and *PPP* 3rd ed. are general reference. Love, *Linux Kernel Development*, sits with the post-core unlocks (§12).

5 The daily session - the unwatched hour

Open the editor, not the book. One kernel, or one optimization step on the current kernel, per session:

- Write the kernel (or the change).
- Verify against the PyTorch reference from the harness.
- Time with CUDA events; compute achieved bandwidth or FLOP rate against the card's spec.
- Profile with ncu; read the limiter section.
- Make one change; re-measure.
- Close with three journal lines: what ran / what limited it / what's next.

The unwatched-hour clause. The session is sized beneath any exam radar: one artifact touched, and every rep ends at a green check visible within minutes - it compiles, it matches the reference, or ncu prints a number. The machines mark; the pen names. Reps are deposits: no rep is ever owed, a gap is represented nowhere, and nothing counts or totals across days. The session's only witness is the profiler, and it witnesses silicon. Exposure material (GPU MODE lectures, kernel blogs) lives in evenings, inside the play thread's time bound - never in the morning slot.

6 The bijection

The loop is timing closure transposed. Hold it that way; the mapping transports predictions, which is the test of a real bijection:

Silicon (owned)	Kernel craft (being bought)
Timing report	Nsight Compute report
What-if analysis on a path	Limiter / occupancy analysis
Critical path	The limiter section
Frequency corners; hard-to-close corner first	Operating regimes (compute- vs memory-bound, batch, dtype); worst regime first
PPA	Latency / throughput / % of speed-of-light
ECO and recovery passes	Micro-optimization iterations
Closure loop: report, hypothesize, change, re-run	Profile loop: ncu, hypothesize, change, re-measure

7 The ladder

One mechanism per rung; the spine text paces it (chapter numbers per the 5th edition). A rung graduates when its kernel is correct, measured, and its limiter is named - facts of the kernel, all three.

#	Kernel	Mechanism trained	PMPP 5e pairing
1	SAXPY / vector add	Launch geometry, indexing, transfers, event timing, bandwidth arithmetic	Ch. 2-3
2	Fused elementwise chain	Kernel fusion, arithmetic intensity, first roofline placement	Ch. 5-6
3	Reduction (sum, then max)	Shared memory, warp shuffles, divergence, atomics	Ch. 10
4	Row-wise softmax	Numerics entry: online max-subtraction, block cooperation - the first numerical-contract kernel	Ch. 10 + numerics appendix
5	Matrix transpose	Coalescing, tiling, bank conflicts	Ch. 5-6
6	Tiled SGEMM: naive, shared-memory tiles, register blocking + vectorized loads	The centerpiece; weeks of reps; where C++ is actually learned. WMMA/tensor-core variant waits for the Thelio	Ch. 3-6 arc; Ch. 23 opens after G2
7	Fused RMSNorm fwd + bwd	Fusion + numerics + gradcheck - the first ML-op artifact (G3's receipt)	Ch. 18 + numerics appendix

Attention forward sits above the ladder, after core. After G1, port the harness itself from Python to C++ (timing, memory management, RAI around CUDA events) - host-side C++ trained organically.

8 Struck until the gates

- Rust.
- Triton - enters after G2 as the productivity layer.
- CUTLASS / CuTe reading - unreadable before G2, illuminating after.
- Leaderboard submissions - in after G2, as the standing scoreboard.
- The PCB place-and-route engine; multi-GPU anything.
- Architecture reading as training - alive only as recreational, time-upper-bounded blog-post research inside the play thread.
- Kinesis Advantage360 typing tests as regular training.
- LeetCode-style maintenance - a small dose near application season, not now. GPU MODE lectures - evenings only, inside the play thread's bound.

9 The receipts rule

Gate receipts are not window dressing. A raw markdown writeup of each gate, pushed to the blog as-is on the day the gate falls - no theme work, no polish, minutes not hours - is core, by two standing laws: publishing is the named antidote to thinking-about-doing, and a thing conceived is not a thing constructed until its verification artifact exists. The post witnesses the kernel - what exists, what it measures, what limits it - never its author. The window dressing rightly deferred is everything decorative around those receipts. A gate without its receipt is not killed; it is remembered.

10 Owner's declared parameters

- Off-Qualcomm undivided attention clocked on its own ledger, separate from the work record.
- Declared budget: about 20 hours/week including weekends; the day opens with this work. (Owner's declaration - this document imposes no clock on it.)

- Qualcomm held at maintenance; the presentation series and senior access are the only floors kept alive there.
- Owner's ruling: progress is recorded in artifact receipts and nothing else - and the receipts measure kernels, never their author.
- Gap rule: returning is everything. A missed day is data; the only entry that matters is the resumption. The return after a day and after a month are identical by construction.

11 The morning - the escort clause

8:30 is the escort's release, never a summons. The night before, the next rep's file is left open on the machine - the morning executes; it does not decide. The vestibule is physical and appetitive, the owner's to design and redesign: out the door before the desk if wanted - light, the walk, coffee, sound - the escort in bodily form. On arrival the axiom's final clause governs and is obeyed, not derived: simply play. Nothing at the desk asks; arriving at one's own pool late costs linear minutes of play and nothing else, by construction.

12 Post-core unlocks

- After G2: Triton; CUTLASS/CuTe reading; leaderboard entries; PMPP Ch. 23 (advanced matrix-multiplication optimizations).
- After all four gates: WMMA/tensor-core work on the Thelio; the attention rung; Love, *Linux Kernel Development*; the portfolio phase - shibboleth kernels first, the writeup genre ("what closing timing teaches about writing kernels"), and the PCB engine only under its standing frontier-attention condition.
- Season: runs on the 2021 A-items - breadth as the floor, warmest channel first, a practice-run target before layer one, a sacrificial first decision interview, detailed recollection after every meeting. The season's calendar lives in the career documents; it never enters the field (§1).

The core closes when G1-G4 have artifact receipts. Rung one is small enough to fall before the coffee cools.